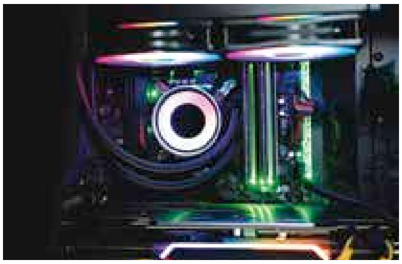
CPU Coolers - Don't just look at socket compatibility

We're all guilty of trying to save a few bucks by compromising on some of the components which we believe don't need a little headroom. Aside from the power supply, the CPU cooler is one such neglected component. When you check the specifications of the CPU coolers to see which all processors are compatible with said cooler, then you'll find even the 120 mm AIO coolers being rated as compatible with the top-of-the-line flagship CPUs. While mechanically the coolers are compatible, there is the question of thermal compatibility. Even if the cooler is thermally compatible, does it come at the cost of running the cooling fans at max RPM even when there's little load? Few companies such as NOCTUA will go the extra mile and even state the thermal compatibility of their coolers but the vast majority only mention physical compatibility. A lot of folks end up buying a Core i9 or Ryzen 9 processor and then pair it with a 120 mm radiator AIO. The end result is that the fans and the water pumps give out well within the warranty period. Before buying an AIO, it's



better to check reviews to see what the load temperatures are for the coolant. Anything that brings the coolant temperature close to 60 degrees Celsius at load should be avoided. 30-40 degrees is what you should aim for at sustained load.

Sound Card - You already have a good one

Honestly, you don't need one. The inbuilt audio solution on most motherboards are more than capable of driving high impedance headsets. Majority of the gaming headsets don't even require an high-end audio solution. Mid-range motherboards these days tend to ship with CODECs that

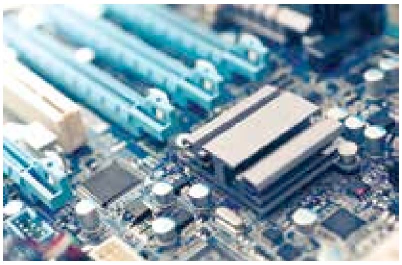
TL;DR

- **CPUs** - More cores isn't necessarily better. The type of cores matters more.
- **Memory** - DDR5 isn't a must have for your next PC build because it offers negligible performance increment.
- **Cooler** - Ensure that you get a cooler that has the ability to dissipate the heat generated by the CPU under load.
- **Sound Card** - The inbuilt solution is more than enough for most people.
- **Motherboard and Power supply** - Refer to expert reviews since there are a lot of gimmicky features.

can drive 250-600 ohms, so they're quite powerful. Where the motherboard audio solution finds itself lacking is when it comes to the choice of ports and interference from other components. If you can hear an audible whine or white noise overlaid onto the sound signals coming out of your PC, then you might have to look towards an alternative sound solution. Usually, the cables are at fault and swapping them out for shielded cables will do the job.

Motherboard - More jargon than everything else

The motherboard is what brings everything together. Folks can get swayed by a lot of marketing jargon around the features that motherboards have. All that excessive information can get quite overwhelming and you might end up buying a motherboard with a lot of features that are really not needed. For example, the power circuitry on motherboards are designed with a ridiculous amount of phases. Processors can easily make do with 6-8 phases using good components. More phases need not necessarily mean that the power supply to the CPU will be more



stable. Things don't work that way all the time. A very common trick motherboard manufacturers implement is "doublers" for the VRM. The manufacturer might claim a higher phase count by adding twice the number of chokes without actually adding proper phases. And since it's not easy for the average joe to discern the number of phases simply by looking at the board, this is another aspect that customers have to figure out by reading expert reviews.

Power Supply - Certification isn't a guarantee

As things get commoditised, manufacturers often resort to throwing in value additions to make their products stand out from the rest of the crowd. Power supplies are undergoing such a phase at the moment. A good power supply should have nice cooling, an efficient AC-DC conversion circuit and an efficient DC-DC power conversion circuit. Everything else is unnecessary. The power efficiency aspect is easily dealt with thanks to the 80Plus rating. It's a certification that's voluntary, however, there are a lot of concerns regarding manufacturers submitting souped up golden samples and getting a high rating but not maintaining the same quality standard for the retail units. This is where PSU Tier Lists come into the picture. Although less scientific, the PSU Tier lists are maintained by independent reviewers who test retail samples and rate their efficiency levels. d